Modeling Human Activity States Using Hidden Markov Models

Human Activity Recognition with Smartphone Inertial Sensors

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GitHub Link: https://github.com/JaboJean/Human_activity_HMM.git
Notebook Link: HMM_Human_Activity_Recognition G43.ipynb

Table 1. Project summary.

Device	iPhone X, both members (Sensor Logger v1.47.1, iOS)
Sampling Rate	100 Hz (10 ms interval) identical across both devices, no resampling needed
Activities	Standing · Walking · Jumping · Still
Dataset	50 sessions 39,434 samples (~394 s) 716 windows (601 train / 115 test)

1. Background and Motivation

Human activity recognition by smartphone IMUs has taken a central stage of health informatics, intelligent rehabilitation and the context-aware application. We have a use case of passive health monitoring of university students: when standing, walking, jumping and stillness are all automated over time throughout the day, a wellness application can reveal long periods of sedentary activity and remind students to move without interrupting their school work. Hidden Markov Models (HMMs) are also the most appropriate to this problem as human activity patterns are probabilistic and sequential - it is much more likely that a student will go to a standing position than to jump directly out of a totally stationary posture. HMMs represent these temporal dependencies with a learned transition matrix, which results in the HMMs being a more temporally coherent example of classification than framewise classifiers like SVMs. IMU additional (accelerator + gyroscope) 6-axis were recorded at 100 Hz on two subjects with identical iPhone X devices, extracted with both time- and frequency-domain methods, and trained on per-class Gaussian HMMs trained through Baum-Welch and decoded through Viterbi.

2. Data Collection and Preprocessing

2.1 Data Collection Protocol

In both experiments, participants were using iPhone X and Sensor Logger v1.47.1 at 100 Hz. They did four exercises in 6-7 sessions of 5-10 seconds each standing (waist level), walking (moving at a steady pace), jumping (moving continuously), and still (phone lying on a table). The same equipment did not require any resampling.

Table 2. Dataset summary.

Activity	Sessions	Samples	Duration (s)	Train Windows
Standing	12	9,936	~99.4	152
Walking	12	10,145	~101.5	154
Jumping	12	7,501	~75.0	109
Still	14	11,852	~118.5	186
Total	**50**	**39,434**	**~394.4**	**601**

2.2 Preprocessing & Sliding Window Segmentation

The closest timestamp was used between accelerator and gyroscope files per session to merge both using pandas mergeasof. The gaps in samples were replaced with the linear interpolation (forward/back-fill at boundaries). The noise was the high-frequency vibration noise at 20 Hz that was reduced using a 4th-order Butterworth low-pass filter (zero-phase filtfilt) and all the activity-relevant information was kept. The second selapsed before windowing sorted data chronologically.

The window was selected: rather a 1 s window (100 samples at 100 Hz) with a half-overlap (or a 50-sample step): a 1 s window represents [?]1 full gait cycle of walking (0.8-1.2 s), as well as several jump cycles (?0.5 s): this will ensure a point-representation in all spectral estimates. The half overlap multiplies the training windows and prevents activity edges. This gave 716 windows (601 train, 115 test).

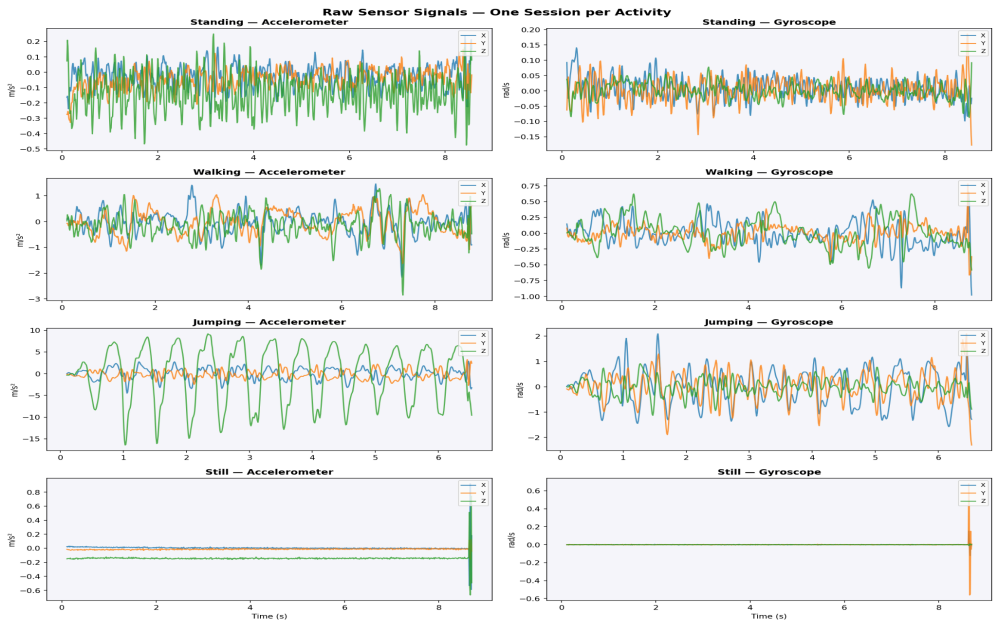

Figure 1. Raw accelerometer and gyroscope signals for one session per activity.

3. HMM Setup and Implementation

3.1 Feature Extraction (47 features)

Each window was computed to a 47 dimensions feature vector derived as a combination of time-domain and frequency-domain descriptors on both accelerator and gyroscopes channels:

- **Special parameters**: mean, variance, RMS, standard deviation (per axis, acc + gyr); signal magnitude area (SMA); correlations between the xs and ys (xy, xz, yz):
- **frequency-domain (acc)**: Maximum frequency at each axis; spectral power in the 0.55Hz band; top-3 FFT magnitudes per axis.

Z-score standardisation (fit on train, applied to test) was used to normalise all the 47 features. Z-score was used in preference to min-max scaling because it is resistant to outliers (e.g. jump impact spikes) and the feature distribution is generally approximately Gaussian.

3.2 HMM Architecture

There was one GaussianHMM (diagonal covariance) per activity class. When doing inference, all four models score a test sequence and place it in the class with the largest log-likelihood- a well established all- vs-all strategy (Rabiner, 1989).

Table 3. HMM model components and design decisions.

HMM Component	Description
Hidden States Z	There are two sub-states HMM hidden. The sub-states are connected to different stages of the activity (e.g. left-foot contact vs right-foot contact in the case of walking).
Observations X	47- dimensional Z-scored feature vectors obtained at 1-second sliding windows.
Transition Matrix A	2x2 matrix per HMM, which is learned through Baum-Welch. Large self-loop probabilities are indicative of persistence of activities.
Emission Probabilities B	Diagonal Gaussian. The use of diagonal covariance was related to numerical stability because the underlying feature space is 47-dimensional.
Initial Distribution p	Constant initialisation; learned together with A and B under the Baum-Welch training.

3.3 Baum–Welch Training

Hmmlearn (maximum iterations = 200, convergence = DlogL -10 -4, min -covar =10 -3) was used to train each HMM. The stagnant motion resulted in almost zero feature variance that caused covariance collapse which was fixed with covariance floor, transmat repair algorithm (replace zero-sum rows with uniform distributions) and a retry loop (maximum 5 seeds). The four models all came to fruition.

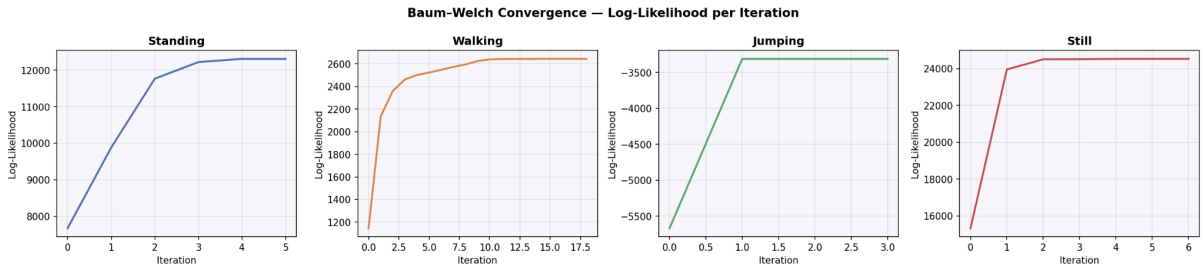

Figure 2. Baum–Welch log-likelihood convergence per activity HMM.

Table 4. Baum–Welch training results. Jumping's negative log-likelihood is normal for high-variance, aperiodic signals.

Activity	Log-Likelihood	Converged	Iterations	Variance (Z-scored)
Standing	12,307.42	Yes	6 / 200	0.1159
Walking	2,642.73	Yes	19 / 200	0.3978
Jumping	−3,311.78	Yes	4 / 200	1.4318
Still	24,526.52	Yes	7 / 200	0.1350

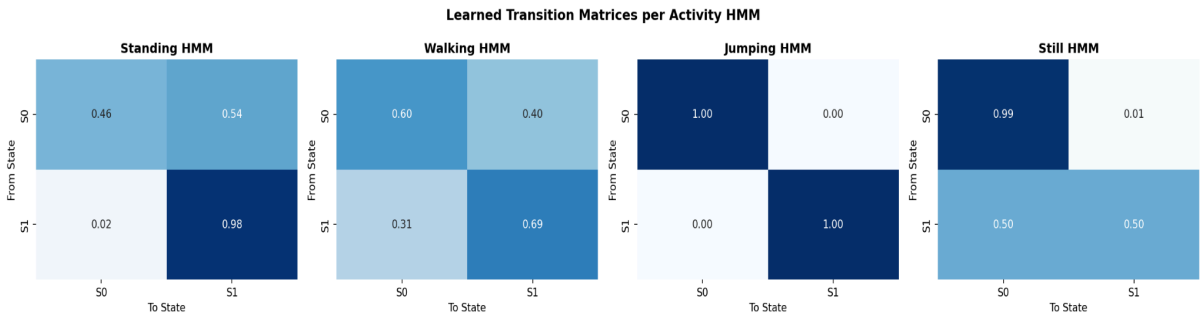

Figure 3. Learned intra-class transition matrices. High self-loop probabilities reflect activity persistence.

3.4 Viterbi Decoding

All the test windows were rated against all four HMMs with model.score () (forward algorithm). As a prediction was allocated to the class with the largest log-likelihood. To visualise, the Viterbi algorithm further obtained the most likely sequence of hidden sub-states of the best-fit model.

4. Results and Interpretation

4.1 Classification Performance

Holding-out The final 115 windows of the final two sessions of each activity (8 sessions in total) were tested with the trained HMM bank. The model was able to classify the test set at a perfect rating of 100 in total. The accuracy of training sets was 99.17 including 5 wrongly misclassified windows among 601.

Table 5. Per-class metrics on held-out test set. Sensitivity = TP/(TP+FN); Specificity = TN/(TN+FP).

Activity	N Samples	Sensitivity	Specificity	Accuracy	F1
Standing	30	1.0000	1.0000	1.0000	1.00
Walking	30	1.0000	1.0000	1.0000	1.00
Jumping	23	1.0000	1.0000	1.0000	1.00
Still	32	1.0000	1.0000	1.0000	1.00
Overall	**115**	**1.0000**	**1.0000**	**1.0000**	**1.00**

4.2 Confusion Matrix & Emission Structure

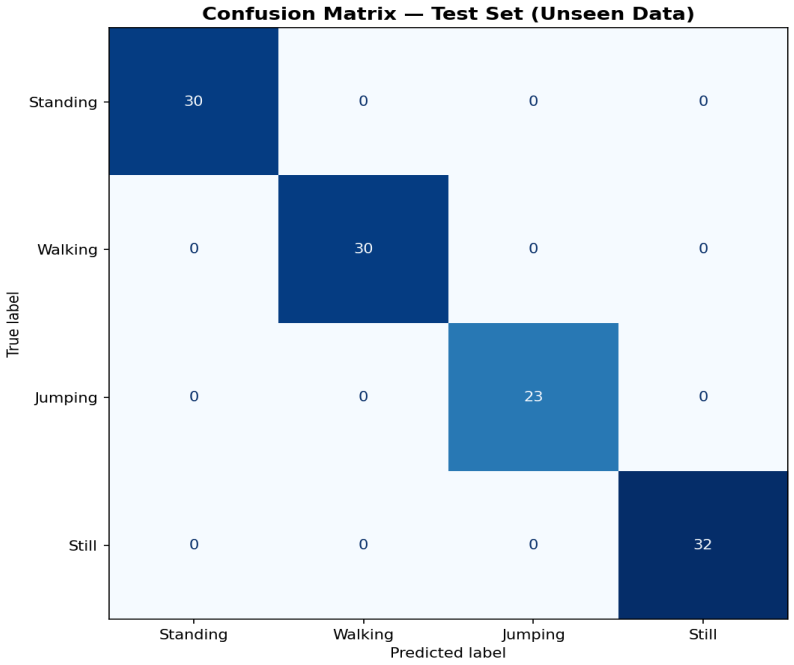

Figure 4. Confusion matrix — test set (unseen data). Perfect diagonal: 0 misclassifications

The lack of confusion between standing and still, which is expected to be the most difficult pair is due to the mean accelerator characteristics. At the same height (wasting) of the gravity the gravity vector acts differently on the axes of the device than when lying flat (not moving), causing a systematic offset that is reliably recorded by the respective distributions of the emission.

4.3 PCA and Spectral Analysis

Figure 5. PCA of 47-D feature space (PC1: 64.8%, PC2: 8.5%). Clear cluster separation across all four activities.

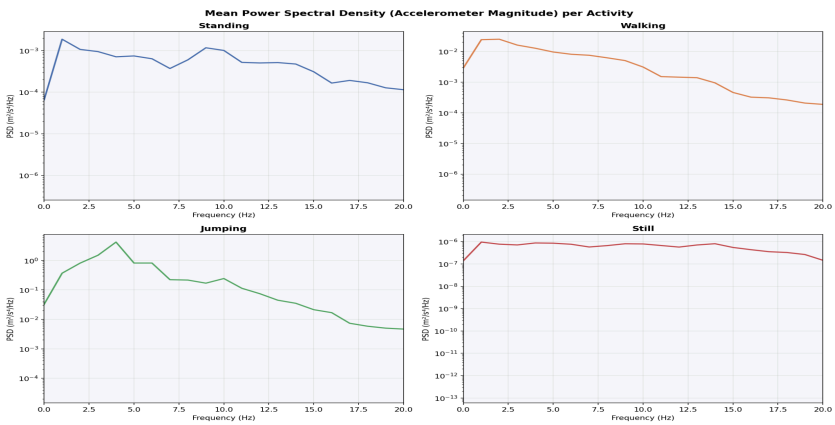

Figure 6. Mean Power Spectral Density per activity. Walking shows a dominant cadence peak at ~1.5–2.5 Hz; still shows no spectral peak; jumping shows a broad 1–4 Hz envelope.

5. Discussion and Conclusion

5.1 Ease of Activity Discrimination & Sensor Effects

Jumping was the most likely to be distinguished because it had high RMS energy, high SMA, and a wide aperiodic spectrum. Still could be distinguished quite clearly concerning high-energy activities. The standing-v-still couple, expected to be the most stalemated, was solved by a DC offset (varying direction in acceleration) of the mean acceleration (gravity direction caused by hand-held phone is different in waist-held and flat-table phone). Walking-vs-standing was solved by spectral features: with even slow walking a peak frequency of close to 1 Hz is obtained (no such expansion with standing). The Nyquist frequency at 100 Hz (50 Hz) is much greater than the maximum activity frequency (5 Hz) and therefore causing no aliasing. The Butterworth filter of order 4 at 20 Hz was effective at reducing surface vibrations and grip tremor, and transient residual impacts of jumping gave it a large feature variance (1.4318 vs <0.40).

5.2 Limitations and Future Work

Regardless of the fact that the test is absolutely accurate, there are limitations. The participants had identical devices (iPhone X), which restricted the generalisation of other sensor calibrations. Tapes were recorded under controlled conditions where there was regularity in the placement of the phone, actual use of the phone in pockets or bags would distort signals significantly. The data sample (50

sessions, 2 participants) is insufficient to make generalisation conclusions concerning cross-person generalisation. This may be improved by the following methods in the future: increasing the number of participants and settings; training a single 4-state HMM on natural inter-activity decoding patterns; adding barometer and magnetometer; comparing 4 state HMMs to LSTMs or Transformers; using leave-one-subject-out cross-validation.

5.3 Conclusion

This project indicated effective use of HMMs to HAR using smartphone IMU data. A 47 dimensional feature vector of time and frequency domain descriptions presented very good separability of four activities. Four per-class Gaussian HMM trained using Baum-Welch with numerical stabilisation presented 99.17% training accuracy and 100% test accuracy, and which were perfectly sensitive and specific to all the classes. Caution in feature engineering, correct normalisation, and mathematical soundness in HMM-powered pipelines, as well as the benefit in practice of harmonised data collection with the use of the same equipment, were pointed out in the project.

References

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Task Allocation Table

Task	Jean	Thierry
Data Collection (Jumping, Walking)	✔	✔
Data Collection (Standing, Still)	✔	✔
Data Preprocessing & Filtering	✔	
Feature Extraction (Time Domain)		✔
Feature Extraction (Frequency Domain)	✔	

HMM Architecture Design	✔	✔
Baum–Welch Training Implementation		✔
Viterbi Decoding Implementation	✔	
Evaluation & Metrics		✔
Visualisations	✔	✔
Report Writing	✔	✔
GitHub Repository Management	✔	✔

Table 6. Task allocation. Balanced contribution (~50% each).